A Shockingly Simple Proof of the Riesz Representation Theorem on Finite-Dimensional Spaces

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1 Abstract

The Riesz Representation Theorem is an important result in linear algebra that allows for the definition of the adjoint. The proof exhibited in this short paper is heavily inspired by Exercise 1-12 in *Calculus on Manifolds* by Michael Spivak [Spi65]. No real mathematical insight has come from the author of this paper, apart from the (very trivial) generalisation of the proof from $\mathbb{R}^n$ to an abstract finite-dimensional inner product space V.

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2 Proof

Riesz Representation Theorem. Let V be a finite-dimensional inner product space, and φ a linear functional on V. Then there exists a unique vector $v \in V$ such that $\varphi(u) = \langle u, v \rangle$, for all $u \in V$.

Proof: For each $v \in V$, define $\psi_v \in V'$ by $\psi_v(u) = \langle u, v \rangle$, for all $u \in V$. Define the linear map $T : V \to V'$ by $Tv = \psi_v$ for all $v \in V$. We claim that T is a bijection from V onto its dual space. Indeed, suppose that $v \in V$ is such that $Tv = 0$. Then in particular, we have the relation

$$\psi_v(u) = \langle u, v \rangle = 0 \tag{1}$$

for all $u \in V$. Choosing $u = v$ in Equation 1, we obtain

$$\psi_v(v) = \|v\|^2 = 0$$

so that $v = 0$ and T is injective as $\ker T = \{0\}$

Since $\dim V = \dim V'$, we have that T is surjective by the Rank-Nullity Theorem. Let $\varphi \in V'$ be given. Our discussion above shows that there exists a unique vector $v \in V$ such that

$$\langle u, v \rangle = [Tv](u) = \varphi(u), \tag{2}$$

for all $u \in V$, which is the statement of the Theorem. ■

References

[Spi65] Michael Spivak. *Calculus on Manifolds*. Mathematics Monograph Series. CRC Press, 1965. ISBN: 9780805390216.